Answer Key

Question	Answer	Question	Answer
1	D	**12**	DELETED
2	A	**13**	B
3	C	**14**	B
4	D	**15**	A , C
5	A	**16**	A, B
6	13	**17**	B
7	0	**18**	A , D
8	2	**19**	30
9	0	**20**	31
10	C	**21**	21
11	A	**22**	04

Converted Solutions

Section A

Q1) Answer = 9

Lets take a general case of two such sequences. (C_i and T_i)

Consider the difference array $d_1, d_2, \ldots, d_{n-1}$, where $d_i = c_{i+1} - c_i$. Let's see what happens if we apply synchronization.

Pick an arbitrary index j and transform c_j to $c'_j = c_{j+1} + c_{j-1} - c_j$. Then:

- $d'_{j-1} = c'_j - c_{j-1} = (c_{j+1} + c_{j-1} - c_j) - cj - 1 = c_{j+1} - c_j = d_j$
- $d'_j = c_{j+1} - c'_j = c_{j+1} - (c_{j+1} + c_{j-1} - c_j) = c_j - cj - 1 = d_{j-1}$

In other words, synchronization simply exchanges two adjacent differences. That means that it's sufficient to check whether the difference arrays of two sets of stones are equal.
For Black Widow, we get : 1 , 2, -5, 6, -3, -2
For Hawkeye, we get : 6, 2, ***x-12 , 10-x,*** -5, -2, 3.
Thus, Case1) **10 - x = 1 and x - 12 = -3** are the two equations obtained from the comparison which gives value of x as **9**. The other case yields x = 13 (Not in options)

Q2) Answer : Yes, No, No, No.

Net cutting heads in each case:

1. 22 - 15 = 7
2. 67 - 25 = 42
3. 1 - 15 = -14
4. 62 - 34 = 28

On observing the number of heads being cut and growing on the dragon(assuming both occur at once) we see that the difference is always a multiple of 7. In each case, consider 4 cases of cutting heads (22,67,1,62 as the final blow) and then we check if we can reduce the dragon heads to that number;

Initial Heads	**Cut 22 in final**	**Cut 67 in final**	**Cut 1 in final**	**Cut 62 in final**
88	66	***21***	87	26
135	113	68	134	73
192	170	125	191	130
152	130	85	151	190

We observe in above table that only 21 is a multiple of 7 and thus the dragon can be killed. (Cutting 22 each time and 15 grow back = total cutting 7) **88 => 81 => 74 => 67** and then cut 67 to kill the dragon.

In all other cases, the dragon can not be killed.

Q3) and Q4)

Considering the height of wall as sequences in each case;

If any two consecutive numbers are having the same parity(parity means even or odd) then remove both the numbers from the sequence. If at the end the sequence has 0 or 1 element then it is possible to do so and in other case it cant be done.
If it is possible then either minimum bricks required is (n*max-sum)/2 is it is integer else (n*(max+1)-sum)/2.
Where max is the maximum number of the sequence.
Sum is the total sum of the sequence.

Q3) Answer : 1456
The numbers are : O, O, O, E, E, O, E, O, O
Total sum of existing wall sizes: 87+451+91+34+234+127+56+9+67 = 1156
Case 1) If the final height of wall is to be 451(Max number in given seq)
{Here dividing by 2 because each brick size is 2}
More bricks required = (451*9 - 1156)/2 = 1451.5 (Not integer).
Case 2) If final height is 452
More bricks required = (452*9 - 1156)/2 = 1456.

Q4) Answer : infinity
The numbers are : E, E, O, O, O, E, E, E. Thus cant be done

Q5) Answer : Shield
First, money spent by each company in that order is 500*0.03 = 15 (This needs to be reduced to find the profits in each case)

From Clue 4 its clear that the Hammer Industries sold at 0.15 and Stark sold at 0.08
From Clue 2, Shield has sold its products at 0.1 and Oscorp at 0.75. Using trail and error, allotting the number of products sold (Keeping Clue 3 in mind), we can find that :
(0.1 * Shield) - 15 = (0.075 * Oscorp) - 15 + 3.8
Shield = 413 with profit 26.30 , Oscorp = 500 with profit 22.50.
(It can be observed that the Shield has more products sale and also the more profit)
From Clue 1, We get that least amount of profit is 26.30 - 11.30 = 15.00, which is satisfied by Stark by selling 375 goods(Observing existing cases).
Mr. Shield sold 413 at $0.1 each for a $26.30 profit .
Ms. Oscorp sold 500 at $0.075, for a profit of $22.50.
Ms. Hammer sold 219 at $0.15 each, for a profit of $17.85.
Mr. Stark sold 375 at $0.08 for a profit of $ 15.00.

Section B

Q7):

Q8):
Ans 2

Solution. Suppose that we have an arrangement satisfying the problem conditions. Divide the board into 2×2 pieces; we call these pieces *blocks*. Each block can contain not more than one king (otherwise these two kings would attack each other); hence, by the pigeonhole principle each block must contain exactly one king.

Now assign to each block a letter T or B if a king is placed in its top or bottom half, respectively. Similarly, assign to each block a letter L or R if a king stands in its left or right half. So we define *T-blocks*, *B-blocks*, *L-blocks*, and *R-blocks*. We also combine the letters; we call a block *a TL-block* if it is simultaneously T-block and L-block. Similarly we define *TR-blocks*, *BL-blocks*, and *BR-blocks*. The arrangement of blocks determines uniquely the arrangement of kings; so in the rest of the solution we consider the 50×50 system of blocks (see Fig. 1). We identify the blocks by their coordinate pairs; the pair (i, j), where $1 \leq i, j \leq 50$, refers to the jth block in the ith row (or the ith block in the jth column). The upper-left block is $(1, 1)$.

The system of blocks has the following properties..

(i′) If (i, j) is a B-block then $(i + 1, j)$ is a B-block: otherwise the kings in these two blocks can take each other. Similarly: if (i, j) is a T-block then $(i - 1, j)$ is a T-block; if (i, j) is an L-block then $(i, j - 1)$ is an L-block; if (i, j) is an R-block then $(i, j + 1)$ is an R-block.

(ii′) Each column contains exactly 25 L-blocks and 25 R-blocks, and each row contains exactly 25 T-blocks and 25 B-blocks. In particular, the total number of L-blocks (or R-blocks, or T-blocks, or B-blocks) is equal to $25 \cdot 50 = 1250$.

Consider any B-block of the form $(1, j)$. By (i′), all blocks in the jth column are B-blocks; so we call such a column *B-column*. By (ii′), we have 25 B-blocks in the first row, so we obtain 25 B-columns. These 25 B-columns contain 1250 B-blocks, hence all blocks in the remaining columns are T-blocks, and we obtain 25 *T-columns*. Similarly, there are exactly 25 *L-rows* and exactly 25 *R-rows*.

Now consider an arbitrary pair of a T-column and a neighboring B-column (columns with numbers j and $j + 1$).

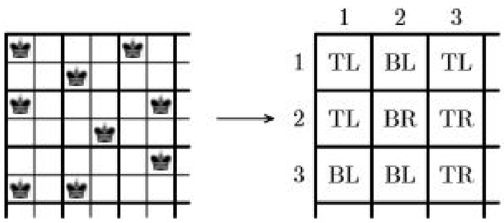


Fig. 1

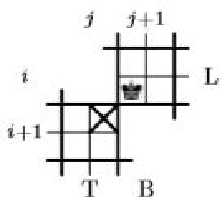


Fig. 2

Case 1. Suppose that the jth column is a T-column, and the $(j + 1)$th column is a B-column. Consider some index i such that the ith row is an L-row; then $(i, j + 1)$ is a BL-block. Therefore, $(i + 1, j)$ cannot be a TR-block (see Fig. 2), hence $(i + 1, j)$ is a TL-block, thus the

$(i+1)$th row is an L-row. Now, choosing the ith row to be the topmost L-row, we successively obtain that all rows from the ith to the 50th are L-rows. Since we have exactly 25 L-rows, it follows that the rows from the 1st to the 25th are R-rows, and the rows from the 26th to the 50th are L-rows.

Now consider the neighboring R-row and L-row (that are the rows with numbers 25 and 26). Replacing in the previous reasoning rows by columns and vice versa, the columns from the 1st to the 25th are T-columns, and the columns from the 26th to the 50th are B-columns. So we have a unique arrangement of blocks that leads to the arrangement of kings satisfying the condition of the problem (see Fig. 3).

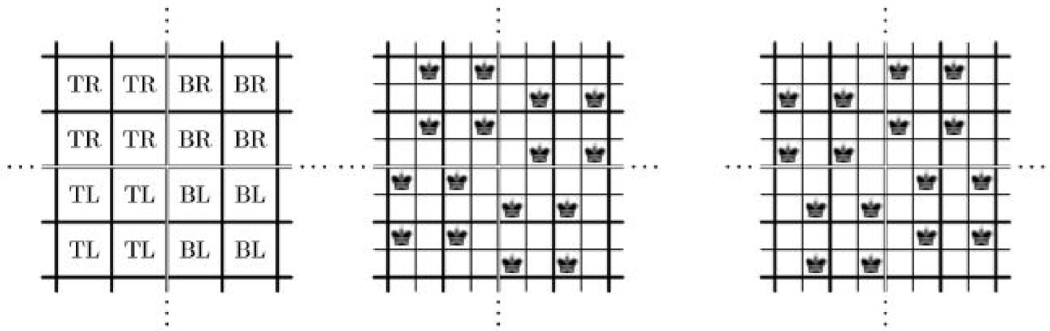


Fig. 3 Fig. 4

Case 2. Suppose that the jth column is a B-column, and the $(j+1)$th column is a T-column. Repeating the arguments from Case 1, we obtain that the rows from the 1st to the 25th are L-rows (and all other rows are R-rows), the columns from the 1st to the 25th are B-columns (and all other columns are T-columns), so we find exactly one more arrangement of kings (see Fig. 4).

Q9):

The instruction of black widow would be reversed exaclty by the opposite instruction of Hawkeye so the delivery boy will comes backs to his initial location so black widow will never win.

Ans) 0

Section C

Q10) and Q11):

Consider the number of spikes in Strange's turn, noting down the winning positions and losing positions:(Win positions are those which can lead to Lose positions of Wong and Losing positions are those which can led to the Win position for Wong in all cases i.e removing 1,3,or 4 spikes)

1. Win (Remove 1 spike)
2. Lose
3. Win (Remove 1 or 3 spikes)
4. Win (Remove 4 spikes)
5. Win (Remove 3 spikes)
6. Win (Remove 4 spikes)
7. Lose and the same sequence repeats from starting.

Q10) Answer : Strange, 1 or 3 spikes

Extending the above sequence till 31, we observe for n=31, it is Win position for Strange by removing 1 or 3 spikes.

Q11) Answer: 49

For less than 50 spikes, 49 is the Lose position for Strange.

Q12) TO BE DELETED

Q13) and Q14)

Let us first consider the first question, as it is significantly easier to answer than the second. Knowing that Rocket is the most accurate and therefore most deadly dueler, we can assume that Rocket would be the first target for Gamora and Drax. Rocket knows that Gamora is more accurate and deadly than Drax, so Rocket would first take aim at Gamora. Furthermore, Drax will not be targeted until either Rocket or Gamora is dead, so it is in Drax's best interest to not interfere with the duel between Rocket and Gamora. In summary, Rocket would aim at Gamora and Gamora at Rocket; Drax would refrain from firing until Rocket or Gamora is killed. Drax is not initially targeted and therefore has the greatest chance of survival.

Q13) Answer : Drax.

Since Drax has the highest chance of surviving, Thanos must have disguised as Drax.

Q14) Answer : 47/90

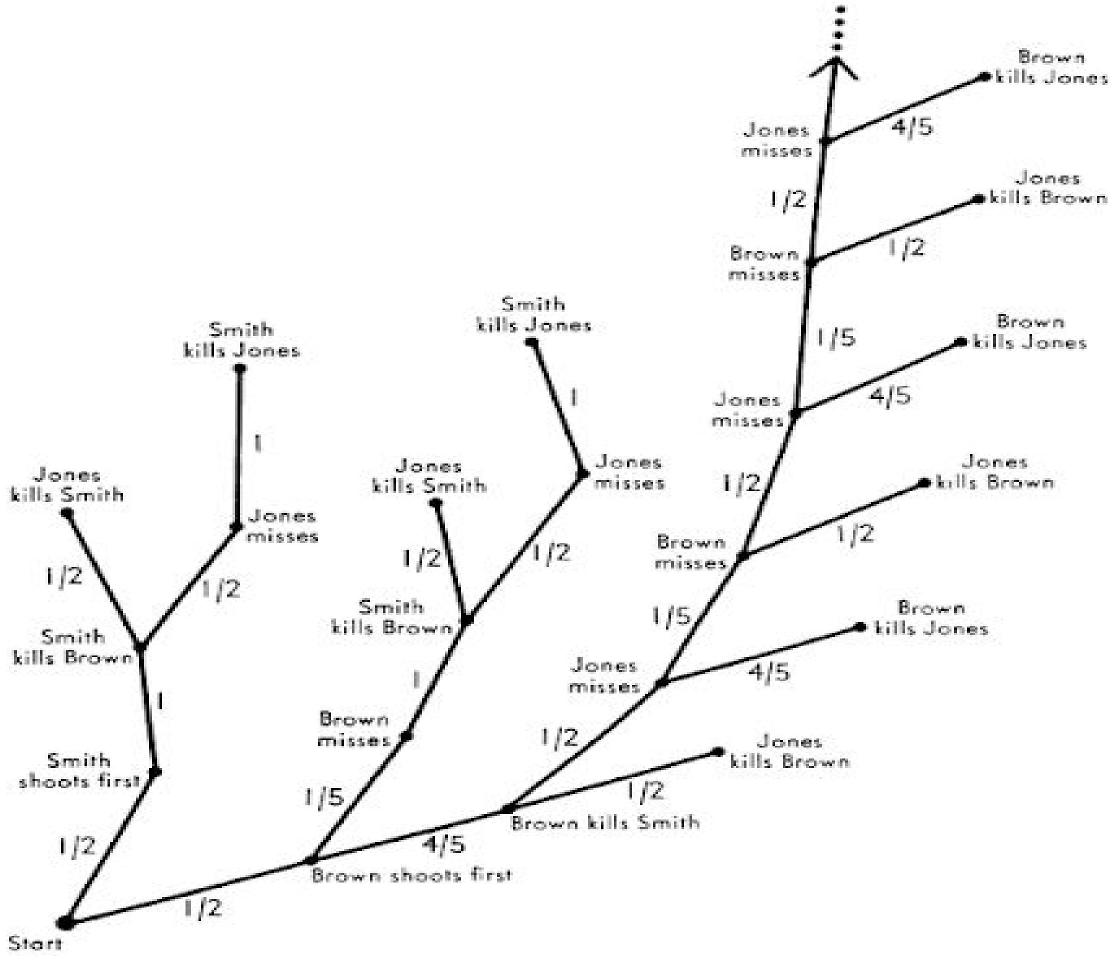
Let's start with Rocket: Rocket has a ½ chance of getting off the first shot in his duel with Gamora (based on who draws a better straw), in which case he will kill Gamora and survive that round. If Gamora shoots before Rocket (a ½ chance), he has a 1/5 chance of missing, so Rocket has a (1/2) * (1/5) chance of surviving. Rocket has a (1/2) + (1/10) = 3/5 chance of surviving the first round of shots. Now Drax will shoot at Rocket, with a ½ chance. If Drax misses, Rocket will shoot him and win. We multiply Rocket's chances for the first round (3/5) with his chances for the second round (1/2) to obtain 3/10, or 30%.

Gamora's case is more complicated because if he wins the first round, he has the potential for an indefinitely long exchange of shots with Drax. Because Rocket's survival chances against Gamora are 3/5, Gamora's chances against Rocket are 2/5 (1 – 3/5). The math gets hairy between Gamora and Drax – I am going to simplify by providing Gardner's insight. The duel between Gamora and Drax becomes a repeating decimal of the series from n = 1 to infinity of 4/(10^n) (for those less mathematically inclined, it looks like 4/10 + 4/100 + 4/1000 . . .). This is equivalent to 4/9. Multiplying Gamora's first and second round chances, we obtain (2/5) * (4/9) = 8/45.

To solve for Drax, we could use the same process or simply subtract the other two probabilities from 1 and obtain 47/90.

Rocket = ⅓; Gamora = 8/45; Drax = 47/90

The entire truel can be conveniently graphed by using the tree diagram shown.



Section D

Q15) Answer : y in first round: Warmachine, z in second round: Vision.

Total sum = 39 points, so we have a sum of either 13 or 3. for 3 it is impossible to find 3 positive integers that sum to 3, so the sum is 13 and they played 3 games.

Consider, hitting x as win, z as lose and y as draw.

The minimum number is at least 1, the maximum number is at most 10. (let's call highest number win, lowest loss, middle draw)

Assume Iron man lost once, then Warmachine can have lost at most twice, and we cannot find such numbers. So Iron man has 2 wins and a draw. Vision and C both have a loss and we still have to split {win,loss,draw,draw} over Vision, Warmachine. Since Vision = Warmachine +1 and Vision won once we have:

Iron man = wwd = > wwl

Vision = llw = > llw

Warmachine = ldd = > ddd

Vision won the last game, so lost the others and Warmachine drawed all the others, so Warmachine also drawed the first one.

Q16) Ans: 0 and 1

Note that the set of sequences of moves the fire makes is in bijective correspondence with the set of strings of Xs and Ys of length 15, where X denotes a move which is either counterclockwise or inward along a spoke and Y denotes a move which is either clockwise or outward along a spoke. (The proof of this basically boils down to the fact that which one depends on whether the fire is on the inner wheel or the outer wheel.) Now the condition that the fire ends at A implies that the difference between the number of Xs and the number of Ys is a multiple of 5, and so we must have either 4, 9 or 14 Xs within the first fourteen moves with the last move being an X . This implies the answer is

14C4+14C9+14C14=3004.
So, the remainders with 2, 3, 7 are : 0, 1, 1 respectively

Q17) and Q18)

The main claim is that if a player is forced to reduce the minimum number of outriders over all piles, then they lose. Intuitively, everytime a player reduces the minimum, the other player has a move that does not reduce the minimum, and if a player isn't forced to reduce the minimum, they have a move that will force the other player to reduce the minimum.
Let m be the minimum number of outriders in a squad and let x be the number of squads with m outriders. Rocket can win if and only if x<=n/2 . Let's call the positions where Rocket can win as winning positions and all other positions as losing positions. To show why this works, we need to show from a winning position, we can reach some losing positions and from a losing position, we can reach only a winning position.
If we are at a winning position, there are at least n/2 squads that have strictly more than m outriders, so we can choose any arbitrary subset of them and reduce them to m outriders. This is now a losing position. If we are at a losing position, no matter what we do, we must include a squad of size m in our chosen subset. If m is zero, this means we have no available moves. Otherwise, the minimum will strictly decrease, but only at pst n/2 squads (from the squads that we chose) can reach that new minimum. Thus losing positions can reach only winning positions. If the frequency of the minimum number is <= n/2, then Rocket wins, otherwise Thor wins.

Q17) Answer: Rocket

The frequency of minimum number “6” is 3, which is less than 5. Thus, Rocket wins.

Q18) Answer : min : 32 and max : 39

If Thor has to land last attack and win, then the frequency of minimum number must be greater than 5. Thus, at least three of a,b,c,d must be 8 and the other number must be >=8 and <=15.
Minimum: 8 + 8 + 8 + 8 = 32
Maximum: 8 + 8 + 8 + 15 = 39

Section E:

Q19) and Q20)

It follows from condition 3 that from 22 through 44 soldiers can be housed. But the number of soldiers must be a multiple of 3 (condition 4). Thus the number can be 24, 27, 30, 33, 36, 39, or 42. Further, trial shows 24 soldiers cannot be housed 11 to a side (condition 3) without leaving empty sections (condition 1), and that 33, 36, 39, or 42 soldiers cannot be housed 11 to a side without placing more than 3 soldiers in some sections (condition 2). By elimination, 30 soldiers were expected, and 27 arrived. The diagrams show how they were housed. (In both pairs of diagrams, the second floor is on the left, and the first floor on the right.)

2	3	3
3	20	2
3	2	2

1	1	1
1	10	2
1	2	1

3	1	3
1	18	2
3	2	3

2	1	1
1	9	1
1	1	1

Q19) Answer: 30

Q20) Answer: 31

M = 2 + 3+ 2+3+1+1+1+1 = 14

N = 3+3+3+3+2+1+1+1 = 17

Q21) Answer: 21

Yes: (1,2),(3, 5), (4, 7), (6, 10), (8, 13), (9, 15), (11, 18), (12, 20), . . .

This series of number pairs is closely related to Fibonacci numbers and the golden ratio. The first pair differ by 1, the second pair by 2, and the nth by n. Every positive integer appears once and only once in the series of pairs. These pairs are called WYTHOFF PAIRS.

Since, max number of stones is 15;

The losing positions for Black Widow are (1,2),(3, 5), (4, 7), (6, 10), (8, 13), (9, 15).

Sum of differences : 1 + 2 + 3 + 4 + 5 + 6 = 21

Q22) Answer: 4

Number the top coin in the pyramid 1, the coins in the next row 2 and 3, and those in the bottom row as 4, 5 and 6. The following four moves are one of the solutions: Move 1 to touch 2 and 4, move 4 to touch 5 and 6, move 5 to touch 1 and 2 below, move 1 to touch 4 and 5.